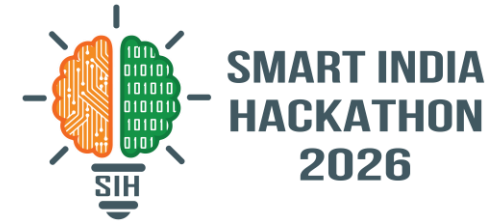


# SMART INDIA HACKATHON 2026



## TITLE PAGE

- Theme – Transportation & Logistics
- PS Category – Software
- Team Name – AAPHAT

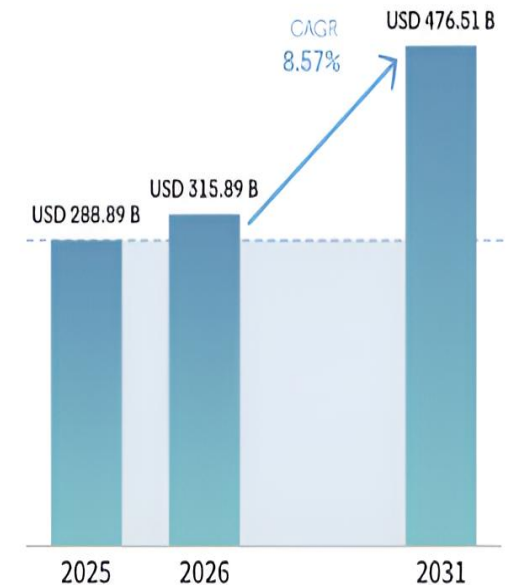




## Proposed Solution

- Two-sided web app: drivers list spare cargo capacity on trips they're already making; senders book a small parcel drop along that route.
- Driver posts route + capacity → customer searches live routes on a map → sends a request → driver accepts/rejects.
- Turns a trucker's wasted return-leg capacity into extra income.
- Gives smaller towns and businesses a cheaper, faster option than standard couriers.
- Unlike BlackBuck/Porter (full business truckloads) or Shadowfax (intra-city), we target one small parcel riding a trucker's own intercity route.
- Live map view + role-based driver/customer dashboards, not just a listing board.

India Freight and Logistics Market  
Market Size in USD Billion



Source: Mordor Intelligence



## Technologies used

- Frontend: React 18 (Vite) + Tailwind CSS + Axios + React-Leaflet maps
- Backend: FastAPI (Python) + SQLAlchemy + JWT auth
- Database: PostgreSQL (Supabase) · Deployed: Vercel + Render

## Methodology & process

- Driver posts a route listing → saved via FastAPI + SQLAlchemy in PostgreSQL
- Customer searches the live map, opens a route, sends a booking request
- Driver accepts/rejects on their dashboard; both sides coordinate pickup
- Every endpoint testable live via FastAPI's auto Swagger docs (/docs)



## Feasibility

- Well-documented, widely-used frameworks (FastAPI, React, PostgreSQL) — realistic for a student team on a hackathon timeline.
- Zero-cost MVP: Vercel + Render + Supabase free tiers cover hosting end-to-end.

## Risks

- 3 separately deployed services (frontend/backend/DB) can cause integration bugs.
- Trust & safety between strangers (driver, customer) is the biggest adoption risk.

## Mitigation strategy

- MVP kept tight: auth, post listing, browse/search, book, accept/reject — chat/payments/admin deferred.
- Swagger docs used to test each backend endpoint before wiring up the frontend.
- OTP verification + ratings flagged as the immediate next milestone.



## Impact on target audience

- Drivers gain a new, trackable income stream from capacity that otherwise travels empty.
- Individuals/small businesses in underserved smaller towns get a cheaper, faster send option.

## Benefits

- Economic: extra driver income; cheaper delivery; commission-based revenue model.
- Environmental: no dedicated extra vehicle trip — rides a journey already happening.
- Social: better delivery access for Tier-2/3 towns & hill regions; formal income for an under-served gig workforce.





## Details / Links of the reference and research work

- [Mordor Intelligence](#)
- [NITI Aayog Freight Optimization Report](#)
- [BlackBuck](#)
- [Porter](#)

